

[illegible]

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/B00T or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull-downs to GND (PROJECT DESIGN CHOICE); reset mode M1/M0=00. AUX is input-only, no pull-down.

OLED: GPIO21/SDA and GPIO22/SCL are signal-only reservations; OLED VCC and I2C pull-ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 -> SN74AHCT1G125DBVR (OE=GND, V5) -> WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE; TPD45311 CC protection is not instantiated: its mandatory 2.7-4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test-point footprints/locations are deferred.

The diagram shows a KICAD schematic for a PCB. It includes various components like connectors (J1, J2, J3), integrated circuits (U1, U2, U3), capacitors (C1-C6), resistors (R8, R9), and test points (TP1-TP7). The schematic is organized into columns labeled 1 through 5. A detailed title block is located in the bottom right corner.

Sheet: /	
File: esp32-e220.kicad_sch	
Title: ESP32 E220 LoRa Receiver – Stage 5 modular carrier	
Size: A4	Date:
KiCad E.D.A. 10.0.5	Rev: Id: 1/1

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/B00T or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

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The schematic diagram illustrates the electrical connections for the Stage 5 modular carrier. It features several key components and interfaces:

- DevKit Interfaces:** J1 and J2 represent USB-C connectors for programming and power. J3 is a universal E220-T22D interface socket.
- Power Management:** The diagram shows various power rails including 5V_SYS, GND, and VBUS_PRE. It includes components like the GRM188R61A106MAJ capacitor and the TPD1E10B06DPYR protection diode.
- Signal Processing:** The E220 module's signals (E220_M0, E220_M1, E220_AUX, E220_RXD, E220_TXD) are connected to the carrier's internal components and test points (TP1-TP7).
- Test Points:** TP1 through TP7 are designated for testing various signals: 5V_SYS, E220_VCC, E220_TXD, M1, AUX, E220_RXD, and E220_TXD respectively.
- Component Footprints:** Various component footprints are shown, including capacitors (C61, C62), resistors (R8, R9), and integrated circuits (U1, U2, U3).

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Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

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TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

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Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

OLED: GPIO21/SDA and GPIO22/SCL are signal–only reservations; OLED VCC and I2C pull–ups are NC/DNP in Rev A.

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Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

OLED: GPIO21/SDA and GPIO22/SCL are signal–only reservations; OLED VCC and I2C pull–ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

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Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

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TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

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Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

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